HEXAWARE

Security Lending Process & FIX Protocol Simulation

Hands-on Training Lab & Modern Capital Markets Reference Standard (2026 Edition)

> **2026 Context:**
>
> • T+1 Settlement & Fast Recall: Securities lending operations operate under strict T+1 settlement rules, requiring real-time automated recall notices and intraday mark-to-market margin calls.
>
> • Messaging & Infrastructure: Upgraded ActiveMQ message queues to Apache Kafka / Event Streams for resilient asynchronous communication between Custodian and Borrower microservices.
>
> • Protocol Standards: Incorporates QuickFIX/J (Java 21 / Spring Boot 3.x) with REST/JSON APIs for hybrid middle-office collateral workflows.
>
> • Institutional Market Leaders: State Street, JPMorgan Chase, BlackRock, Northern Trust, BNY Mellon, and BNY Mellon Agency Securities Lending.

1. Introduction to Securities Lending

Securities Lending involves the temporary transfer of securities (shares, sovereign bonds, or corporate debt) from an institutional lender to a borrower, secured by collateral (cash, government securities, or high-grade equities). The borrower pays a monthly fee or rebate rate for the loan and remains contractually obligated to return the exact securities on demand or at loan maturity.

Primary lenders in global capital markets include large asset owners and asset managers such as mutual funds, pension funds, insurance companies, and sovereign wealth funds. Primary borrowers include broker-dealers, hedge funds, market makers, and proprietary trading desks.

1.1 Key Motivations & Market Mechanics

Lender Motivations (Supply Side):

- • Portfolio Yield Enhancement: Offsets portfolio management and custody expenses by generating incremental fee revenue.
- • Collateral Reinvestment: Reinvests cash collateral in short-term money market instruments to earn reinvestment spread.

Borrower Motivations (Demand Side):

- • Covering Short Positions: Fulfills settlement obligations for short sales in compliance with regulatory fail-delivery rules.
- • Arbitrage & Hedging: Facilitates convertible arbitrage, dividend arbitrage, and complex multi-asset hedging strategies.

2. Industry Architecture & Market Participants

In institutional markets, security lending is typically administered through Agent Lenders (custodial banks) who administer lending programs, negotiate rebate rates, manage collateral haircuts, and process corporate actions.

3. Scope of Work (Trainee Application Architecture)

Trainees will design and implement a dual-microservice application simulating a Securities Lending transaction between a Custodian (Agent Lender) and a Borrower (Hedge Fund / Prime Broker) using Java 21, Spring Boot 3.x, and Apache Kafka.

3.1 FIX Messaging Workflow Sequence

Seq #	Message Name	FIX Type	Interaction Sequence & Business Description
1	Request For Security	AF	Custodian -> Borrower: Custodian initiates a availability/loan request (AF) specifying required security symbol and quantity.
2	Collateral Inquiry	BB	Borrower -> Custodian: Borrower sends Collateral Inquiry (BB) to request required collateral haircut and eligible collateral types.
3	Collateral Report	BA	Custodian -> Borrower: Custodian returns Collateral Report (BA) detailing eligible assets, margins, and valuation requirements.
4	Trade Capture Report	AE	Borrower -> Custodian: Borrower submits Trade Capture Report (AE) containing agreed loan quantity, fee rate, and settlement date.
5	Trade Confirmation	35=AK	Custodian -> Borrower: Custodian verifies MSLA limits and issues Trade Confirmation (AK) locking the loan contract.
6	Security Status	f	Borrower -> Custodian: Borrower broadcasts Security Status (f) updating loan execution status (e.g., Allocated, On-Loan, Recalled).

4. Microservice Personas & Functional Requirements

4.1 Custodian Microservice (Agent Lender)

- • User Interface: Interactive dashboard displaying active Master Securities Lending Agreements (MSLA), open loan contracts, collateral balance coverage, and incoming borrow requests.
- • Backend Logic: Listens for incoming Kafka events, validates counterparty credit limits, formats and transmits AF, BA, and AK messages.

4.2 Borrower Microservice (Prime Broker / Hedge Fund)

- • User Interface: Portfolio borrow portal enabling traders to initiate borrow requests against custodians, review collateral ratios, and track locate requests.
- • Backend Logic: Formats and publishes BB, AE, and 'f' messages to Kafka, updating local loan inventory databases upon confirmation.

5. Trainee Practical Coding Assignment

> **Technical Deliverables:**
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> 1. Microservice Setup: Develop two independent Spring Boot 3.x services (custodian-service and borrower-service) using Java 21.
>
> 2. Event Streaming: Configure Apache Kafka topics ('sec-lending-requests', 'sec-lending-responses') for asynchronous message exchange.
>
> 3. Database Integration: Maintain trade state in PostgreSQL/H2 tables tracking TransactionID, LoanQty, RebateFee, CollateralValue, and ContractStatus.
>
> 4. T+1 Recall Enhancement: Add an intraday Recall Notice endpoint allowing the Custodian to issue a recall message for early loan termination under T+1 settlement constraints.